

Quantum Dice Trinity Challenge 2026

Resource Allocation and Job Scheduling in Agentic AI

1. Problem Context

Modern agentic AI systems consist of multiple specialised agents performing tasks such as reasoning, retrieval, coding, planning, or tool execution.

As these systems scale, a central challenge emerges:

How should jobs be assigned across agents and compute resources efficiently under operational constraints?

In practice, multiple jobs compete for limited resources such as compute, memory, bandwidth, or specialised agents. Certain jobs may require specific capabilities, while others may conflict in time or resource usage.

Examples include:

- Multi-agent orchestration systems
- Distributed inference scheduling
- AI workflow execution
- Autonomous task routing
- Resource-aware LLM deployment

These systems naturally form combinatorial optimisation problems involving discrete assignment decisions, resource contention, and scheduling constraints.

In this challenge, you are asked to formulate such a problem in a way that can be solved using a probabilistic computing framework. The emphasis is on how constraints, conflicts, and resource limitations are represented within the optimisation problem.

2. Mathematical Starting Point

Consider a set of jobs indexed by $i = 1, \dots, N$ and a set of agents indexed by $a = 1, \dots, A$.

Introduce binary decision variables:

$$x_{i,a} \in \{0, 1\} \tag{1}$$

where $x_{i,a} = 1$ if job i is assigned to agent a .

A basic constraint is that each job is assigned once:

$$\sum_{a=1}^A x_{i,a} = 1 \tag{2}$$

Agents may have limited compute or memory capacity. Jobs may also conflict in time or compete for the same resources.

For any conflicting pair (i, j) :

$$x_{i,a} + x_{j,a} \leq 1 \quad (3)$$

The objective is typically to minimise some operational cost:

$$\min \sum_{i,a} c_{i,a} x_{i,a} \quad (4)$$

where $c_{i,a}$ may capture factors such as latency, compute usage, communication overhead, or execution preference.

To move into a form suitable for probabilistic optimisation, constraints are incorporated into a quadratic objective:

$$\min_{\mathbf{x}} \mathbf{x}^T Q \mathbf{x} \quad (5)$$

where Q encodes both assignment costs and penalties for violating constraints.

3. Modelling the Real Problem

The core difficulty lies in how constraints and interactions are represented.

Assignment constraints introduce structured couplings between variables associated with a given job. Resource and scheduling conflicts introduce additional interactions across jobs and agents.

Depending on the application, additional considerations may arise. These may include:

- Agent capability matching
- Resource capacity limits
- Prioritisation of urgent jobs
- Workflow dependencies
- Communication costs between agents
- Dynamic or online scheduling

Deciding which constraints are hard and which are soft, and how strongly they are enforced, is a central modelling decision.

Different formulations may produce very different optimisation landscapes and solution behaviour.

Participants should view the formulation itself as something to be designed and explored.

4. Computational Perspective

This class of problems is closely related to scheduling, graph colouring, bin packing, and task allocation problems.

The number of variables scales with the number of jobs and agents, while the number of interactions grows with the density of constraints and conflicts.

As systems become more congested, the optimisation landscape becomes increasingly coupled and difficult to navigate.

From a computational perspective, the structure of the formulation itself strongly affects tractability, scalability, and solution quality.

Participants are encouraged to think carefully about:

- Constraint structure
- Interaction sparsity
- Penalty weighting
- Scalability of the formulation

5. References

The following references provide useful entry points into scheduling and combinatorial optimisation formulations:

- M. L. Fisher, “The Lagrangian Relaxation Method for Solving Integer Programming Problems,” *Management Science*, 1981.
- A. Lucas, “Ising formulations of many NP problems,” *Frontiers in Physics*, 2014.
- M. Pinedo, *Scheduling: Theory, Algorithms, and Systems*.
- E. G. Coffman et al., “Approximation Algorithms for Bin Packing,” *Approximation Algorithms for NP-Hard Problems*, 1997.